

```
In [ ]: import matplotlib.pyplot as plt  
import seaborn as sns
```

```
In [ ]: tips = sns.load_dataset("tips")
```

```
In [ ]: tips.head()
```

Line plot

Line plots are used to visualize trends in data over time or other continuous variables. In a line plot, each data point is connected by a line, creating a smooth curve. In Seaborn, line plots can be created using the `lineplot()` function.

```
In [ ]: sns.lineplot(x="total_bill", y="tip", data=tips)
```

Bar plot

Bar plots are used to visualize the relationship between a categorical variable and a continuous variable. In a bar plot, each bar represents the mean or median (or any aggregation) of the continuous variable for each category. In Seaborn, bar plots can be created using the `barplot()` function.

```
In [ ]: sns.barplot(x="sex", y="total_bill", data=tips)
```

Scatter plot

Line plots are used to visualize trends in data over time or other continuous variables. In a line plot, each data point is connected by a line, creating a smooth curve. In Seaborn, line plots can be created using the `lineplot()` function.

```
In [ ]: sns.scatterplot(x="total_bill", y="tip", data=tips)
```

Histogram

Histograms visualize the distribution of a continuous variable. In a histogram, the data is divided into bins and the height of each bin represents the frequency or count of data points within that bin. In Seaborn, histograms can be created using the `histplot()` function.

```
In [ ]: sns.histplot(x="tip", data=tips)
```

```
In [ ]: sns.scatterplot(x="total_bill", y="tip", data=tips)
```

Density Plot

Density plots, also known as kernel density plots, are a type of data visualization that display the distribution of a continuous variable. They are similar to histograms, but instead of representing the data as bars, density plots use a smooth curve to estimate the density of the data. In Seaborn, density plots can be created using the `kdeplot()` function.

```
In [ ]: sns.kdeplot(data=tips, x="total_bill")
```

Heatmap

A heatmap is a graphical representation of data that uses colors to depict the value of a variable in a two-dimensional space. Heatmaps are commonly used to visualize the correlation between different variables in a dataset.

```
In [ ]: corr = tips.corr()

sns.heatmap(corr)
```

Set figure size

```
In [ ]: plt.figure(figsize=(12, 8))

corr = tips.corr()

sns.heatmap(corr)

plt.show()
```

Set plot title, xlabel, ylabel

```
In [ ]: # Create a density plot of the "total_bill" column from the "tips" dataset

# We use the "hue" parameter to differentiate between "lunch" and "dinner" meal times

# We use the "fill" parameter to fill the area under the curve

# We adjust the "alpha" and "linewidth" parameters to make the plot more visible

# Set figure size
plt.figure(figsize=(12, 8))

# Plot density of total_bill for lunch and dinner respectively
sns.kdeplot(data=tips, x="total_bill", hue="time", fill=True, alpha=0.6, linewidth=2)

# Add a title and labels to the plot using Matplotlib

plt.title("Density Plot of Total Bill by Meal Time")

plt.xlabel("Total Bill ($)")

plt.ylabel("Density")

# Show the plot

plt.show()
```

Create subplots

```
In [ ]: fig, ax = plt.subplots(nrows=1, ncols=2, figsize=(18, 6))
sns.barplot(x='sex', y='total_bill', data=tips, ax=ax[0])
ax[0].set_title('Total Bill by Sex', fontsize=16)

sns.barplot(x='day', y='total_bill', data=tips, ax=ax[1]);
ax[1].set_title('Total Bill by Day of Week', fontsize=16)

plt.show()
```

```
In [ ]:
```